MIT 18.06

Linear Algebra

Lecture Notes

1 ~ 5

Lecture 1 The Geometry of Linear Equations

Linear Algebra: Solve a system of linear equations
n equations, n unknown

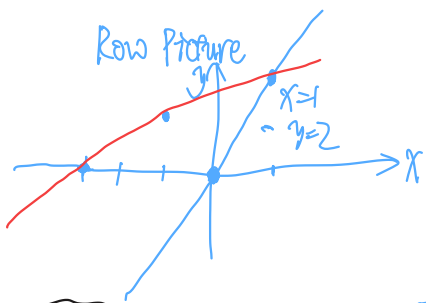
Example ①

$$2x - y = 0$$
$$-x + 2y = 3$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$A \quad x = b$$

Row Picture

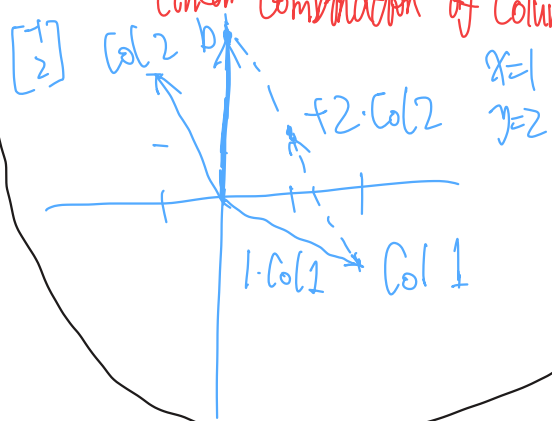


Linear Combination of Columns

Column Picture

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix},$$

Linear Combination of Columns

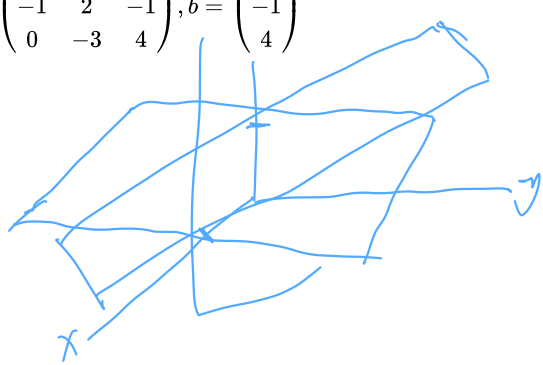


Example ②

$$2x - y = 0$$
$$-x + 2y - z = -1$$
$$-3y + 4z = 4$$

Row Picture

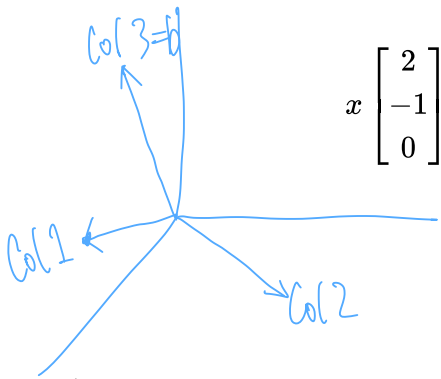
$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{pmatrix}, b = \begin{pmatrix} 0 \\ -1 \\ 4 \end{pmatrix}$$



not clear

Column Picture

$$x\begin{bmatrix}2\\-1\\0\end{bmatrix}+y\begin{bmatrix}-1\\2\\3\end{bmatrix}+z\begin{bmatrix}0\\-1\\4\end{bmatrix}=\begin{bmatrix}0\\-1\\4\end{bmatrix}\qquad x=0, y=0, z=1$$



$$x\begin{bmatrix}2\\-1\\0\end{bmatrix}+y\begin{bmatrix}-1\\2\\3\end{bmatrix}+z\begin{bmatrix}0\\-1\\4\end{bmatrix}=\begin{bmatrix}1\\1\\3\end{bmatrix}$$

$$x=1, y=1, z=0$$

- Can I solve $Ax=b$ for every b?
- Do the linear combinations of the columns fill 3D space?
  - For this A, the answer is YES
  - NO: if column 3 is a combination of col 1 and col 2
    - not invertible

fill the space

- Imagine a 9-D space, 9 equations, 9 unknowns
  - if col 8 = col 9, we can only fill a 8-D plane in 9-D space
- $Ax=b$

$$\begin{bmatrix}2&5\\1&3\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix}=1\begin{bmatrix}2\\1\end{bmatrix}+2\begin{bmatrix}5\\3\end{bmatrix}=\begin{bmatrix}12\\7\end{bmatrix}$$

Ax is a linear combination of the columns of A

Lecture 2 Elimination with Matrices

$$x + 2y + z = 2$$
$$3x + 8y + z = 12$$
$$4y + z = 2$$

$Ax = b$

Elimination - Elimination < success / failure

$$\underbrace{\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}}_{A} \xrightarrow{(2,1)} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} \xrightarrow{(3,2)} \underbrace{\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}}_{U}$$

first pivot

second pivot

upper triangle

- How could this have failed?
  - if first pivot = 0, if 8 is 6, if last 1 is -4

Back-substitution - Back-substitution

$$\underbrace{\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix}}_{b} \xrightarrow{(2,1)} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} \xrightarrow{(3,2)} \underbrace{\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 2 \\ 6 \\ -10 \end{bmatrix}}_{c}$$

$Ux = c$

$$x + 2y + z = 2 \qquad x = 2$$
$$2y - 2z = 6 \longrightarrow y = 1$$
$$5z = -10 \qquad z = -2$$

solving the system in reverse order because the system is triangular.

$$\begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{matrix} 3\times \text{col}1 \\ + \\ 4\times \text{col}2 \\ + \\ 5\times \text{col}3 \end{matrix}$$

linear combination of columns

Matrix × column = column

$$\underset{1\times 3}{\begin{bmatrix} 1 & 2 & 7 \end{bmatrix}}\underset{3\times 3}{\begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}} = \begin{matrix} 1\times \text{row}1 \\ +2\times \text{row}2 \\ +7\times \text{row}3 \end{matrix}$$

linear combination of rows

Elimination Matrices

- Matrices: subtract $3\times$ row 1 from row 2

$$E_{21}\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

- Step 2: subtract $2\times$ row2 from row 3

$$E_{32}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}$$

$$E_{32}(E_{21}A) = U \rightarrow (E_{32}E_{21})A = U$$

Permutation

- Permutation: exchange row 1 and 2

$$P\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

- Row operations
  Matrix multiples on the left

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix}$$

- Column operations
  Matrix multiples on the right

Inverses

Inverses

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E^{-1}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{I}$$

add 3 times row 1 to row 2

subtract 3 times row 1 from row 2

Lecture 3 Multiplication and Inverse Matrices

Multiplication

$$\begin{array}{c} \\ \text{row } 3 \end{array} \begin{bmatrix} \\ a_{31} \; a_{32} \; ___ \end{bmatrix} \begin{bmatrix} & \text{col } 4 \\ & b_{14} \\ & b_{24} \\ & \\ & \end{bmatrix} = \begin{bmatrix} \\ c_{34} \\ \\ \end{bmatrix}$$

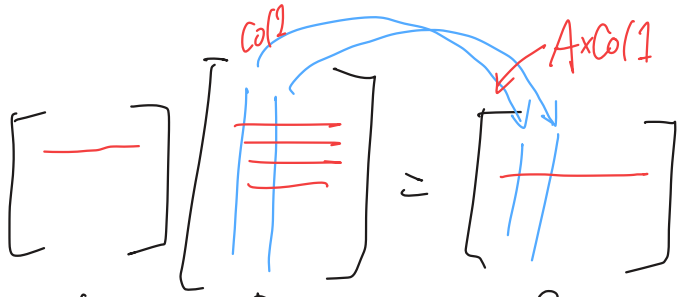
$A_{m \times n}$ $\quad B_{n \times p}$ $\quad C = AB_{m \times p}$

$$c_{34} = (\text{row } 3 \text{ of } A) \cdot (\text{col } 4 \text{ of } B)$$

$$= a_{31} b_{14} + a_{32} b_{24} + \dots$$

$$= \sum_{k=1}^{n} a_{3k} b_{k4}$$

regular way



$A_{m\times n}$ $B_{n\times p}$ $C_{m\times p}$

Columns of C are combinations of columns of A.

- column of A × row of B (?)

$$\begin{bmatrix}2\\3\\4\end{bmatrix}^{m\times 1}\begin{bmatrix}1 & 6\end{bmatrix}^{1\times p} = \begin{bmatrix}2 & 12\\3 & 18\\4 & 24\end{bmatrix}^{m\times p}$$

rows of C are combinations of rows of B

4th way

- 4th way: AB = Sum of (cols of A)(rows of B)

$$\begin{bmatrix}2 & 7\\3 & 8\\4 & 9\end{bmatrix}\begin{bmatrix}1 & 6\\0 & 0\end{bmatrix} = \begin{bmatrix}2\\3\\4\end{bmatrix}\begin{bmatrix}1 & 6\end{bmatrix} + \begin{bmatrix}7\\8\\9\end{bmatrix}\begin{bmatrix}0 & 0\end{bmatrix}$$

Block

- Block

$$\underbrace{\left[\begin{array}{c|c}A_1 & A_2\\ \hline A_3 & A_4\end{array}\right]}_{A}\underbrace{\left[\begin{array}{c|c}B_1 & B_2\\ \hline B_3 & B_4\end{array}\right]}_{B} = \left[\begin{array}{c|c}A_1B_1+A_2B_3 & \\ \hline & \end{array}\right]$$

Inverse

- Inverses (Square Matrices)

$$A^{-1}A = I = AA^{-1}$$

if A^{-1} exists

invertible, nonsingular

Singular case

- Singular case (no inverse)

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$$

If you can find a vector x with $Ax = 0$ $\quad x \neq 0$

$$Ax = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Gauss-Jordan Elimination

- Gauss-Jordan Elimination (solve 2 equations at once)

$$\underbrace{\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} a & c \\ b & d \end{bmatrix}}_{A^{-1}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{I}$$

$A \times$ column j of A^{-1} = column j of I

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{array}\right]$$

$A \quad I \qquad\qquad I \quad A^{-1}$

$$\underline{\underline{E}} \begin{bmatrix} A & I \end{bmatrix} = \begin{bmatrix} I & A^{-1} \end{bmatrix}$$

E's

$EA = I$ tells us $E = A^{-1}$

Lecture 4 Inverse of AB, A^T, Factorization into $A=LU$

Inverse of AB, A^T

$$AA^{-1} = I = A^{-1}A$$

$$(AB)(B^{-1}A^{-1}) = I \qquad B^{-1}A^{-1}AB = I$$

$$AA^{-1} = I \qquad (A^{-1})^T A^T = I$$

This is $(A^T)^{-1}$, Inverse of A^T

$A=LU$

$$\underbrace{\begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix}}_{E_{21}} \underbrace{\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}}_{U}$$

$$E_{32}E_{31}E_{21}A = U \quad \text{(no row exchanges)}$$

$$\underbrace{\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}}_{U}$$

$$A = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U$$

$$\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$$

$$A = LU$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -5 & 1 \end{bmatrix}}_{E_{32}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{21}} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 10 & -5 & 1 \end{bmatrix} = E \text{ (left of } A)$$

$$EA = U$$

Inverses (Reverse order)

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix} = L \text{ (left of } U)$$

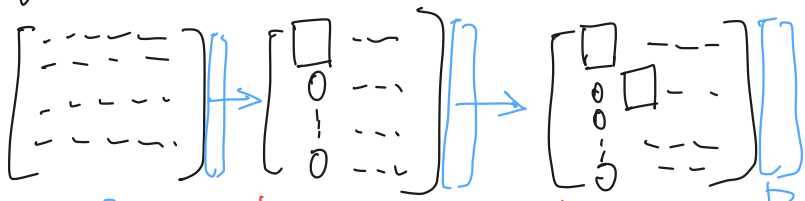
$$A = LU$$

$A=LU$ if no row exchanges, multipliers go directly into L

Cost

How many operations (= multiply + subtract) on $n \times n$ matrix A?

$n = 100$



cost? $n, n^2, n^3, n!$ about 100^2 about 99^2 B

count: $n^2 + (n-1)^2 + \cdots + 1^2 \approx \frac{1}{3} n^3$ on A

Calculus is like sums except it's continuous, where Algebra is discrete. Cost of B is n^2

Permutations

Permutations 3×3 — 6 P's $P^{-1} = P^T$ 4×4 — 24 P's

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Lecture 5 Transposes, Permutations, Spaces R^n

Permutations

Permutations P: execute row exchanges

$$A = LU = \begin{bmatrix} 1 & & & 0 \\ - & 1 & & \\ - & - & \ddots & \\ - & - & - & 1 \end{bmatrix} \begin{bmatrix} \diagdown & - & - \\ 0 & \diagdown & - \\ & & \diagdown \end{bmatrix}$$

$\hookrightarrow$ Becomes $PA = LU$ // any invertible A

permutations P = identity matrix with reordered rows

$n! = n(n-1)\cdots(3)(2)(1)$ counts reorderings

counts all $n \times n$ permutations, $P^{-1} = P^T$, $P^T P = I$

Transpose

Transpose

$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}$$

R^T $\quad$ R

$$(A^T)_{ij} = A_{ji}$$

Symmetric matrices

Symmetric matrices $A^T = A$

Example $\begin{bmatrix} 3 & 1 & 7 \\ 1 & 2 & 9 \\ 7 & 9 & 4 \end{bmatrix}$

$R^T R$ is always symmetric

$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 11 & 7 \\ 11 & - & - \\ 7 & - & - \end{bmatrix}$$

Why?

Take transpose

$(R^T R)^T = R^T R^{TT} = R^T R$

Vector Spaces

Vector Spaces

Examples: R^2 = all 2-dim real vectors $\begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \pi \\ e \end{bmatrix}$

= "x-y plane"

R^3 = all vectors with 3 real components $\begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$

R^n = all column vectors with n real components

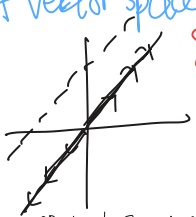
Not a vector space

It is not closed.

A vectors space has to be closed under $\times, +$ linear combinations.

Subspace

A vector space inside R^2



line in R^2 through zero vector

Subspace of R^2

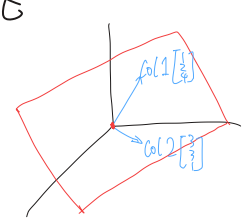
1. all of R^2
2. any line through $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
3. zero vector only Z

subspace of R^3

1. R^3
2. plane
3. line
4. Z

$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

column space

$A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$ columns in R^3

all their combinations form a subspace called column space $C(A)$